VCD-170522 FY B.Sc Computer Science-Sem II **Calculus** **2.30 hrs – 75 marks**

Q.1 Attempt all multiple choice questions: **[40 marks]**

A) A function is said to be invertible if and only if it is
a) injective
b) surjective
c) bijective
d) constant function

B) Gradient of $f(x,y) = x^3 + 2xy^2$ at (1,1) is
a) (3,4)
b) (5,4)
c) (4,3)
d) (4,5)

C) $\int_1^2 3x^2dx$ is
a) 6
b) 7
c) 8
d) 9

D) $\int 2^x dx =$
a) $2^x + c$
b) $2^x log2 + c$
c) $2^x + log2 + c$
d) $\frac{2^x}{log2} + c$

E) Integrating factor of differential equation $\frac{dy}{dx} + \frac{1}{x}y = 1$ is
a) 1
b) 1/x
c) x
d) logx

F) Area under the curve $y = x^2$ over the interval [-1,1] is
a) 4
b) 1/2
c) 1/3
d) 2/3

G) Solution of differential equation $\frac{dy}{dx} = -xy$ is
a) $\frac{y^2}{2} = \frac{x^2}{2} + c$
b) $logy = \frac{x^2}{2} + c$
c) $\frac{-y^2}{2} = \frac{x^2}{2} + c$
d) $logy = \frac{-x^2}{2} + c$

H) Unit vector of 3i + 4j is
a) (3i + 4j)/5
b) (3i + 4j)/7
c) 3i + 4j
d) (3i + 4j)/12

I) $f(x,y) = ysinx - e^x$, then f_{yx} is
a) cosx
b) -cosx
c) sinx
d) -sinx

J) $\lim_{(x,y)\to(2,2)} \frac{x^2-xy}{x-y}$ is
a) 0
b) 1
c) 2
d) does not exist

110522-

K) Which of the following is not true?

a) $\nabla(f+g) = \nabla f + \nabla g$
b) $\nabla(f-g) = \nabla f - \nabla g$
c) $\nabla(f.g) = g.\nabla f + f.\nabla g$
d) $\nabla(f/g) = g.\nabla f - f.\nabla g$

L) There is no change in function f at point u in the direction of vector v if

a) $\hat{v}$ *is in the direction of* ∇f
b) $\hat{v}$ *is in the direction of* $-\nabla f$
c) $\hat{v}$ *is perpendicular to* ∇f
d) $\hat{v}$ makes angle $\frac{\pi}{3}$ *with* ∇f

M) $\int x\sin x\, dx$ is

a) $x\cos x + \sin x + c$
b) $x\cos x - \sin x + c$
c) $-x\cos x + \sin x + c$
d) $-x\cos x - \sin x + c$

N) The interval in which the function $f(x) = x^2$ is increasing is

a) $(0, \infty)$
b) $(-\infty, 0)$
c) $(0,2)$
d) $(-2,0)$

O) Rate of change of one variable with respect to another is called

a) continuity
b) *integral*
c) derivative
d) inflexion

P) Critical point of the function $f(x) = x^2 + 6x$ is

a) -6
b) -3
c) 6
d) 3

Q) $\int \frac{1}{x} dx$ is

a) $\log|x| + c$
b) $\frac{x^2}{2} + c$
c) $\frac{2}{x^2} + c$
d) $\frac{-1}{x^2} + c$

R) Absolute maximum value of f(x) = $x^2 - 4x$ in [1, 3] is

a) -1
b) -2
c) -3
d) -4

S) A function f(x) is concave up on an interval I if for all $x \in I$

a) $f'(x) > 0$
b) $f'(x) < 0$
c) $f''(x) > 0$
d) $f''(x) < 0$

T) Derivative of e^2 is

a) e^2
b) 0
c) $\frac{e^3}{3}$
d) 2e

Q.2) Attempt the following (solve any 02) **[10 marks]**

i) Divide 100 into two parts such that sum of their squares is minimum.

ii) Draw the graph of the curve $y = 4 - 3x^2 + x^3$.

iii) Find the intervals in which the function $f(x) = 2x^3 - 12x^2 + 18x + 15$ is increasing or decreasing.

iv) Find the horizontal and vertical asymptotes of the function $f(x) = \frac{x}{(x-2)(x+3)}$.

Q.3) Attempt the following (solve any 02) **[10 marks]**

i) Use Euler's method with a step size of 0.25 to find approximate solution of initial value problem $\frac{dy}{dx} = x + 2y$, y(0)=0 over $0 \le x \le 1$.

ii) Estimate $\int_0^4 x^2 dx$ using Simpson's rule where n = 4.

iii) Solve the differential equation: $\frac{dy}{dx} - y = 2x$.

iv) Find the area bounded between the graph of cosx, sinx and y-axis on $[0, \pi/4]$.

Q.4) Attempt the following (solve any 02) **[10 marks]**

i) Find second order derivatives of the function $f(x,y) = x^3y^2 - 3x^2y + y^3 + 1$

ii) Find the local extrema or saddle points of the function $f(x,y) = 3x^2 - 2xy + y^2 - 8y$

iii) Find the directional derivative of the function $f(x,y) = x^3 + 2xy^2$ at the point (-2, -3) in the direction of the vector $\vec{a} = i + j$.

iv) Find the equation for the tangent plane and parametric equation for normal line to the surface $x^2 + 4y = z^2$ at the point (1,4,3).

Q.5) Attempt the following (solve any 01) **[05 marks]**

i) Find the absolute maximum and minimum values of the function $f(x) = x^2 - 4x + 2$ in [0,3].

ii) Solve the differential equation: $\frac{dy}{dx} = -4xy^2$

iii) Let $z = x^2y, x = t^2 \text{ and } y = t^3$. Use chain rule to find $\frac{dz}{dt}$.